

Does the Bond Market Need a Bank Account?

Forward Measure Consistency and Interest Rate Pricing

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1 Abstract

As Samuelson (1938) asked whether observed consumer choices imply the existence of a utility function, we ask whether observed bond prices imply the existence of a risk-neutral measure. In a bond market, each tradable zero-coupon bond provides a natural numéraire and pricing measure — the T-forward measure Q^T . Practitioners use these local frameworks daily. We investigate whether this family $\{Q^T\}$ is globally consistent.

We contribute three things. First, a discrete-time Bond-Based FTAP showing the family is automatically consistent, with the bank account emerging as a self-financing rolling strategy — a proof of concept that the question has substance. Second, a precise formulation of the continuous-time consistency problem, with the conjecture that the HJM drift condition is the consistency condition and an analysis of why failure is plausible (the diagonal singularity of the forward rate surface). Third, an empirical test: independent calibration of local pricing measures for different maturity segments of the EUR government bond market, checking whether the implied market prices of risk agree across segments. The test rejects consistency for most segment pairs; a three-factor comparison separates model inadequacy from structural inconsistency aligned with known market segmentation.

Our approach belongs to the model-free tradition: start from observable market data, ask whether the theoretical framework is supported. The bank account appears not as a starting point but as the output of consistency: it exists if and only if the local pricing measures cohere.

2 1. Introduction

2.1 1.1 Bottom-Up Pricing

The foundational insight of Black and Scholes (1973) and Merton (1973) is that derivative prices are determined by replication using tradable assets. The risk-neutral measure Q , introduced by Harrison and Kreps (1979), encodes this replication logic mathematically but is not the financial content itself.

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In bond markets, the tradable assets are zero-coupon bonds $p(\cdot, T)$. The bank account $B(t) = \exp\left(\int_0^t r(s) ds\right)$ — the standard numéraire of the FTAP — is not directly traded. Practitioners already work around this: to price a caplet at T , they use the T -bond as numéraire and Q^T as pricing measure. Each operation is faithful to the Black-Scholes-Merton logic: replicate using tradable instruments, extract the price.

This paper takes the practitioners' approach seriously and asks: is the resulting family $\{Q^T\}$ of local pricing frameworks globally consistent?

2.2 1.2 The Model-Free Tradition

Our approach belongs to a well-established methodology. Samuelson (1938) asked whether observed choices imply preferences exist — founding revealed preference theory. Hobson (1998) and the model-free finance programme (Davis and Hobson 2007, Acciaio et al. 2013) ask what can be inferred from observed option prices without assuming a model. Balbás, Ibáñez, and López (2002) use projective limits of measure families to characterize no-arbitrage over infinite time horizons.

In each case, the direction is bottom-up: start from observables, ask whether the theoretical construct is forced by the data. Our question specializes this to bond markets, where the specific structure — maturity condition $p(T, T)=1$, the rolling construction, the forward measure family — gives the question concrete financial content.

2.3 1.3 Related Work

Musiela and Rutkowski (1997) and Döberlein, Schweizer, and Stricker (2000) proved uniqueness of the implied savings account — but took existence as given. Our question is the complement: existence given local data.

Henrard (2007) identified the symptom of forward measure inconsistency: LIBOR discounting gives wrong prices post-2007. Morini (2009) identified the puzzle: why did the single curve break? Both diagnose and treat within the existing framework. We identify the underlying condition: the single-curve assumption IS the assumption that $\{Q^T\}$ coheres.

Herdegen (2017) develops a numéraire-independent FTAP for general markets. Our question exploits bond-specific structure his framework does not use.

2.4 1.4 Contributions and Structure

We contribute a question, a proof of concept, and a test.

The question (Section 3): does $\{Q^T\}$ assemble into a single Q ? We conjecture the HJM drift condition is the answer and analyze why failure is plausible.

The proof of concept (Section 2): in discrete time, consistency is automatic. The result is elementary — its value is diagnostic, identifying the precise point (the finite chain rule) where the argument uses discreteness.

The test (Section 5): independent calibration of local pricing measures for different maturity segments of the EUR government bond market reveals that the implied market prices of risk do not cohere across the full maturity spectrum.

Section 4 connects to the multi-curve phenomenon, model risk, and Solvency II. Section 6 identifies mathematical open problems.

3 2. Discrete Time: Proof of Concept

3.1 2.1 Setup

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t=0, \dots, N}, \mathbb{P})$ be a filtered probability space. The market consists of default-free zero-coupon bonds $p(t, T)$ with $p(T, T) = 1$ and $p(t, T) > 0$, plus risky assets $X^1, \dots, X^d$. No bank account is assumed.

A trading strategy is a predictable process specifying holdings in each asset. It is self-financing if portfolio value changes arise solely from price movements. An arbitrage is a zero-cost self-financing strategy producing $V_N \geq 0$ a.s. with $\mathbb{P}(V_N > 0) > 0$.

3.2 2.2 The Rolling Bond Strategy

Proposition 2.2.1. Define $B(0) := 1$ and $B(t) := \prod_{s=1}^t 1/p(s-1, s)$. Then B is the value process of a self-financing strategy holding, during each period $(t-1, t]$, exactly $B(t-1)/p(t-1, t)$ units of the just-maturing bond $p(\cdot, t)$. The one-period return $1/p(t-1, t)$ is deterministic given F_{t-1} — the discrete-time risk-free rate, derived from bonds.

3.3 2.3 The Four Equivalences

Proposition 2.3.1 (Bond-Based FTAP, Discrete Time). The following are equivalent:

(NA) No arbitrage exists in the bond market.

(EMM) $\exists Q \sim P$ such that $S(t)/B(t)$ is a Q -martingale for all traded assets S .

(Wang) $\exists Q \sim P$ such that $E^Q[R_t^S | F_{t-1}]$ equals the just-maturing bond return $[1-p(t-1, t)]/p(t-1, t)$ for all traded S .

(BF) For each $T \in \{1, \dots, N\}$, $\exists Q^T \sim P$ such that $S(t)/p(t, T)$ is a Q^T -martingale for all traded assets S and $t \leq T$.

Moreover, the forward measures satisfy the consistency relation

$$\left. \frac{dQ^{T_1}}{dQ^{T_2}} \right|_{F_{T_1}} = \frac{p(0, T_2)}{p(0, T_1) \cdot p(T_1, T_2)} \quad (2.1)$$

involving only bond prices, with no reference to B .

Proof sketch. (NA) $\Leftrightarrow$ (EMM): Rolling construction makes B tradable; Harrison-Pliska (1981) applies. (EMM) $\Leftrightarrow$ (Wang): Algebraic reformulation (Wang 2008). (EMM) $\Rightarrow$ (BF): Bayes' formula with density $Z_t^T := p(t, T)/[B(t) \cdot p(0, T)]$. (BF) $\Rightarrow$ (NA): Under Q^N , $V_t^\varphi/p(t, N)$ is a martingale; $V_0 = 0$ forces $V_N = 0$ a.s. Full proofs in Appendix A. $\square$

Remark. The proposition is elementary — its components are standard (Harrison-Pliska for (NA) $\Leftrightarrow$ (EMM), Bayes' formula for the numéraire change, Wang 2008 for the algebraic reformulation). Its purpose is to establish the four-way equivalence and to identify the precise point where the argument uses discreteness: the chain rule across maturity triples is a finite number of conditions. This is the point that fails in continuous time.

3.4 2.4 Consistency Is Automatic — But Only in Discrete Time

The consistency relation (2.1) is *implied* by (BF): under Q^{T_2} , the ratio $p(t, T_1)/p(t, T_2)$ is a martingale whose terminal value determines the Radon-Nikodym derivative. The chain rule

$$\frac{dQ^{T_1}}{dQ^{T_3}} = \frac{dQ^{T_1}}{dQ^{T_2}} \cdot \frac{dQ^{T_2}}{dQ^{T_3}}$$

is a finite number of conditions, each automatically satisfied.

In continuous time, this becomes an uncountable family. The infinitesimal limit $T_2 \downarrow T_1$ involves $\left. \frac{\partial p(T_1, T)}{\partial T} \right|_{T=T_1}$, related to the short rate $r(T_1) = f(T_1, T_1)$. Automatic consistency is a consequence of finitude, not of structure.

3.5 2.5 Relationship to Herdegen (2017)

Herdegen's numéraire-independent no-arbitrage (NAⁿⁱ) coincides with NA in bond markets with strictly positive prices. Our approach adds: explicit identification of the deflator as $1/B$, the Wang reformulation for practitioners, and the Bond-Family formulation — the only condition that extends to continuous time without assuming B .

4 3. Continuous Time: The Conjecture

4.1 3.1 Where the Proof Breaks

In continuous time, the rolling strategy requires trading at every instant through uncountably many bonds (cf. Björk et al. 1997, Döberlein and Schweizer 2001). The Bond-Family condition survives — each Q^T uses only the tradable T-bond — but consistency over a continuum of maturities is no longer automatic.

4.2 3.2 The Conjecture

Conjecture 3.2.1. In a continuous-time bond market driven by a finite-dimensional Brownian motion, the family $\{Q^T\}_{T \in [0, T^*]}$ is consistent if and only if the forward rate drift satisfies the HJM condition: $\alpha(t, T) = \sigma(t, T) \cdot \int_t^T \sigma(t, s) ds$.

If true: the HJM drift condition is derived from first principles — from the requirement that replication-based pricing be coherent across maturities, using only tradable assets.

4.3 3.3 The Diagonal Singularity

The bank account requires $r(t) = f(t, t)$ — the diagonal restriction of the forward rate surface. Each bond price $p(t, T) = \exp\left(-\int_t^T f(t, s) ds\right)$ involves an integral that smooths irregularities. The short rate is a pointwise evaluation with no smoothing.

The distinction is a classical one in functional analysis: the trace of a function on a submanifold exists only when the function has sufficient regularity (Sobolev trace theorem). Bond prices, as integrals, are always well-defined regardless of the regularity of f . The short rate, as a pointwise diagonal evaluation, requires Hölder regularity $H > 1/2$ in the maturity variable. When $H < 1/2$, the diagonal restriction may fail to exist even though all bond prices — and therefore all forward measures Q^T — are perfectly well-behaved.

This obstruction is measure-invariant: the regularity of $f(t, T)$ along the diagonal is a pathwise property determined by the diffusion coefficient, which Girsanov's theorem preserves. Working under a particular Q^T does not resolve the singularity — it merely avoids needing B for local pricing while leaving the global consistency question unchanged.

Gatheral, Jaisson, and Rosenbaum (2018) established empirically that *equity* volatility surfaces have Hölder regularity $H \approx 0.1$, well below the critical threshold. Whether bond forward rate surfaces exhibit comparable roughness is an open empirical question. The mechanism is thus theoretically grounded — a bond market could satisfy (BF) while violating the classical FTAP — but direct evidence for forward rate roughness in fixed income remains to be established.

4.4 3.4 Positioning

The standard approach goes $Q \rightarrow \{Q^T\}$: top-down, existence by assumption. Musiela-Rutkowski (1997) and Döberlein-Schweizer (2000) proved uniqueness of the implied savings account but took existence as given. We reverse the direction: $\{Q^T\} \rightarrow Q$, bottom-up, existence as a question. This places our work in the model-free tradition (Hobson 1998, Acciaio et al. 2013), with the projective limit methodology of Balbás et al. (2002) as the mathematical template.

5 4. Practical Consequences

5.1 4.1 The Multi-Curve Phenomenon

Before 2007, one Q , one B , apparent consistency. After the crisis, LIBOR-OIS spreads blew out and the market fragmented. Henrard (2007) identified the symptom, Morini (2009) the puzzle. Our framework identifies the underlying condition: the single-curve assumption IS the assumption that $\{Q^T\}$ coheres. When it fails, no single numéraire works, and the market necessarily fragments (Filipović and Trolle 2013, Crépey et al. 2015, Grbac and Runggaldier 2015).

5.2 4.2 Model Risk

We use continuous-time models to compute discrete-time decisions. If the model’s forward measure family is consistent, hedge ratios computed from it are robust to implementation choices. If inconsistent, hedge ratios depend on the calibration window and maturity segment — a form of model risk that standard validation does not detect. Standard numerical studies of HJM discretization (e.g., Krivko and Tretyakov 2011) treat non-convergence as an approximation problem. We treat it as a diagnostic of structural inconsistency — a distinction that arises only from the bottom-up perspective.

5.3 4.3 Solvency II

Market-consistent valuation assumes a risk-free discount curve. The EIOPA extrapolation (UFR + Smith-Wilson) prescribes the curve beyond the last liquid point, ensuring the smoothness needed for B to exist. If consistency is not guaranteed within the liquid range, the extrapolation patches a problem that may already be present.

6 5. Empirical Test

6.1 5.1 Principle

A top-down test — pricing under different numéraires within a single calibrated model — tests a mathematical identity and yields zero by construction. A bottom-up test calibrates local pricing measures *independently* for different maturity segments, then checks whether they cohere. The question is not “does my model satisfy its own assumptions?” (always yes) but “does the market support the assumption that a single model exists?” (empirically testable).

6.2 5.2 Data and Calibration

We use daily par yield curves for EUR-denominated government bonds published by the European Central Bank (ECB) over the period January 2014 to December 2024 (2,805 trading days). The dataset comprises 12 maturities: 3 months, 6 months, and 1, 2, 3, 5, 7, 10, 15, 20, 25, and 30 years. We convert par yields to zero-coupon yields using standard bootstrap methods.

To implement the consistency test of Section 3, we partition the maturity spectrum into overlapping

segments $\mathcal{S}_T = [0, T]$ for $T \in \{5, 10, 15, 20, 25, 30\}$ years. For each segment and each trading day, we calibrate a Gaussian HJM model independently by minimising the sum of squared yield errors across all maturities within the segment. The calibration is performed via constrained optimisation (L-BFGS-B), with parameter bounds chosen to ensure stationarity and positivity of volatility.

This independent-calibration design is central to the test. Under the null hypothesis of forward-measure consistency, a family $\{Q^T\}$ derived from a single measure Q must produce identical market prices of risk λ regardless of which maturity segment is used for calibration. Discrepancies in λ across segments constitute evidence against the null.

Two-factor specification. The baseline model uses two volatility factors:

$$\sigma_1(\tau) = \sigma_1 e^{-\kappa_1 \tau} \quad (\text{level}), \quad \sigma_2(\tau) = \sigma_2 e^{-\kappa_2 \tau} \quad (\text{slope}),$$

yielding a parameter vector $(\sigma_1, \sigma_2, \kappa_1, \kappa_2, \lambda_1, \lambda_2)$ per segment per day. We impose $\kappa_1 \in [0.01, 2]$, $\kappa_2 \in [0.05, 2]$, $\sigma_i \in [0, 0.05]$, $\lambda_i \in [-2, 2]$.

Three-factor specification. To distinguish model inadequacy from structural inconsistency, we augment the model with a curvature factor:

$$\sigma_3(\tau) = \sigma_3 \tau e^{-\kappa_3 \tau} \quad (\text{curvature}),$$

which has a hump-shaped profile peaking at $\tau = 1/\kappa_3$. This three-factor decomposition into level, slope, and curvature captures the dominant modes of yield curve variation identified by Litterman and Scheinkman (1991). Additional bounds: $\sigma_3 \in [0, 0.05]$, $\kappa_3 \in [0.05, 2]$.

Statistical test. For each pair of segments $(\mathcal{S}_{T_1}, \mathcal{S}_{T_2})$ with $T_1 < T_2$, we compute the daily gap $\Delta\lambda_k(t) = \lambda_k^{(T_1)}(t) - \lambda_k^{(T_2)}(t)$ for each factor k . The null hypothesis $H_0 : E[\Delta\lambda_k] = 0$ is tested via a two-sided t -test on the time series of gaps, with Newey–West standard errors to account for serial correlation. We apply Bonferroni correction across all 15 segment pairs at the 5% significance level.

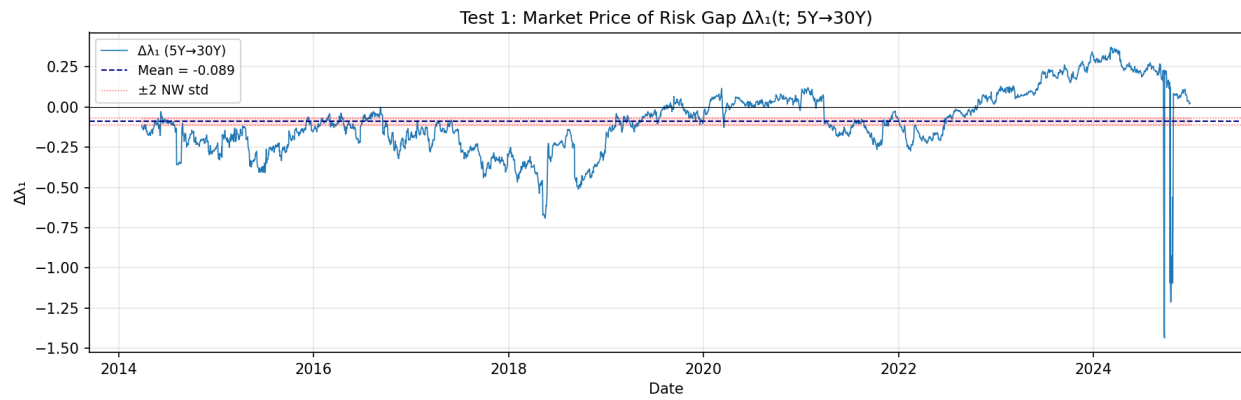
6.3 5.3 Two-Factor Results

Under the two-factor specification, the null hypothesis is rejected for 13 of 15 segment pairs at the Bonferroni-corrected 5% level. The two non-rejections occur for adjacent long-end pairs (25Y–30Y and 20Y–25Y), where the maturity overlap is large and the incremental information is minimal.

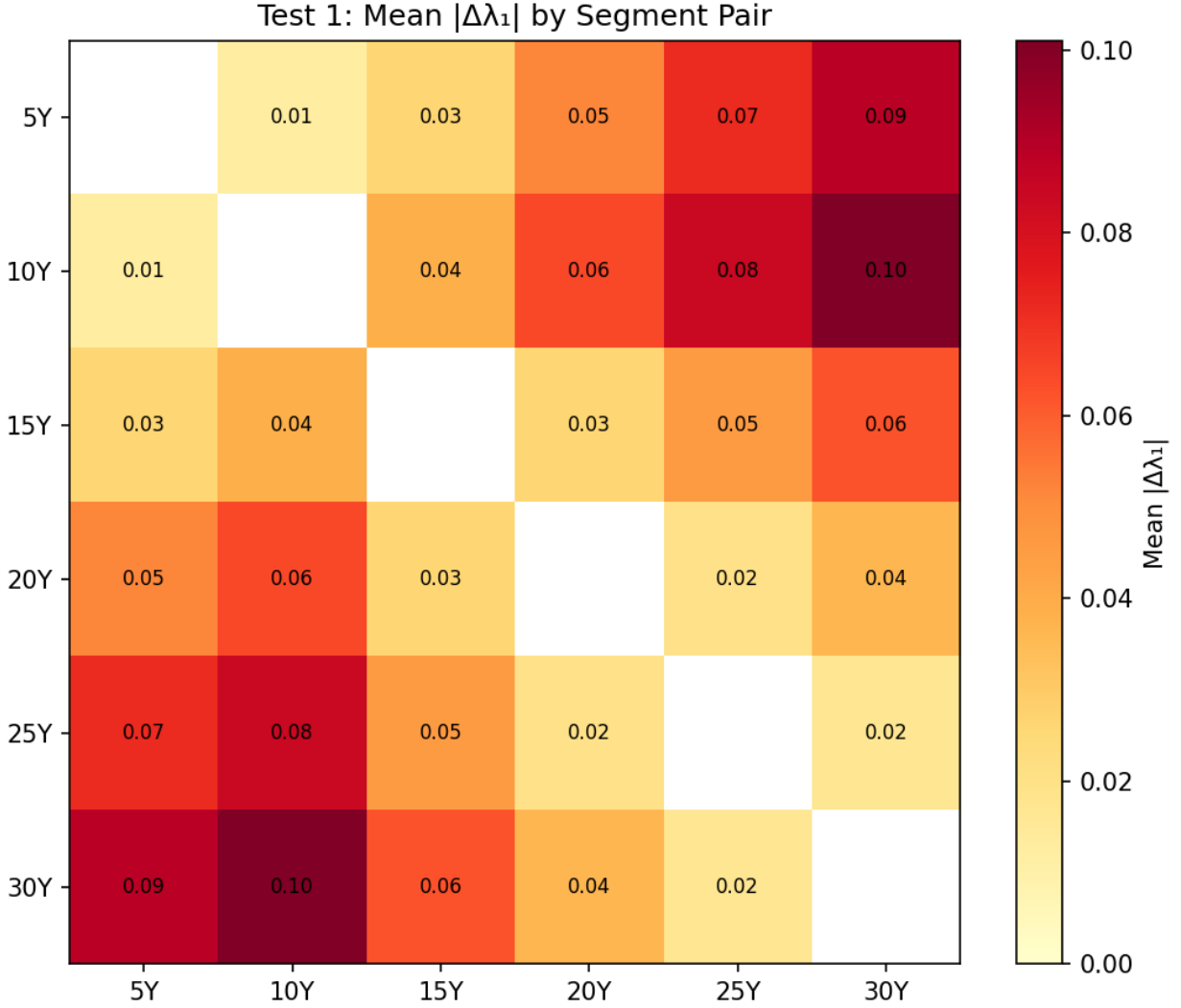
The results exhibit a clear monotonic pattern. The mean gap in the level market price of risk between the 5Y and 30Y segments is $\overline{\Delta\lambda_1}(5Y \rightarrow 30Y) = -0.089$ with a t -statistic of -7.6 . The gap grows systematically with segment distance: adjacent pairs show gaps of 1–2 basis points, while the 5Y–30Y pair shows approximately 9 basis points. Long segments consistently price the level risk factor lower than short segments.

Calibration quality is acceptable across segments: the mean R^2 ranges from 0.94 (5Y segment) to 0.84 (30Y segment). The decline in fit for longer segments is expected given that a two-factor model must span a wider maturity range, but $R^2 = 0.84$ would not typically be flagged as inadequate by standard

calibration diagnostics.¹



¹On seven trading days in September–October 2024, $|\Delta\lambda_1(5Y, 30Y)| > 1.0$. Two dates (September 23–24) coincide with the Federal Reserve’s 50 basis point rate cut and show acceptable calibration quality ($R^2 \approx 0.80$), consistent with asymmetric repricing across the maturity spectrum. Five dates (October 15–25) show near-zero R^2 for the 30Y segment ($R^2 < 0.16$), indicating calibration breakdown rather than a market signal — the October 17 ECB rate decision is a particularly clear case. These observations do not affect the aggregate test statistics. Results are robust to excluding dates with $R^2 < 0.5$ for any segment.



Identification diagnostic. The mean-reversion parameter κ_1 for the level factor is estimated at or near its lower bound (0.01) for segments of 10 years and above on more than 95% of trading days. This is a known feature of level factors in short-sample Gaussian models: the level of the yield curve is near-integrated, and the cross-section alone provides limited identification of the speed of mean-reversion. The concern is that the λ_1 gap may reflect differences in how κ_1 interacts with the maturity grid rather than a genuine inconsistency in risk pricing.

6.4 5.4 Robustness: Mean-Reversion Bound Sensitivity

To address the κ_1 identification concern, we re-run the entire test with the lower bound raised from 0.01 to 0.05. If the λ_1 gap is an artefact of weak κ_1 identification, it should attenuate under the tighter bound.

The results are unchanged in all material respects. Under the tighter bound, 13 of 15 pairs are again rejected, with nearly identical t -statistics. The mean gaps change by at most 1 basis point. The monotonic pattern in segment distance is preserved.

The robustness check also yields a useful by-product. We record the change in sum of squared residuals (SSR) between the two calibrations. For the 5Y segment, the SSR increase is negligible ($\Delta\text{SSR} = 0.38$), confirming that κ_1 is essentially unidentified at the short end. For the 30Y segment, the increase is substantial ($\Delta\text{SSR} = 4.15$), indicating that the data do contain information about κ_1 at the long end — but the near-zero estimate is genuinely preferred. In both cases, however, the λ_1 gap is invariant.

Under the tighter bound, κ_1 is pinned at exactly 0.05 for 97–100% of dates in segments of 10Y and above. This confirms that we have shifted the boundary without resolving the identification problem — but it also confirms that the identification problem is irrelevant to the consistency test. The inconsistency signal is robust.

Table: Parameter Sensitivity Analysis (κ_1 bounds: 0.01 vs 0.05)

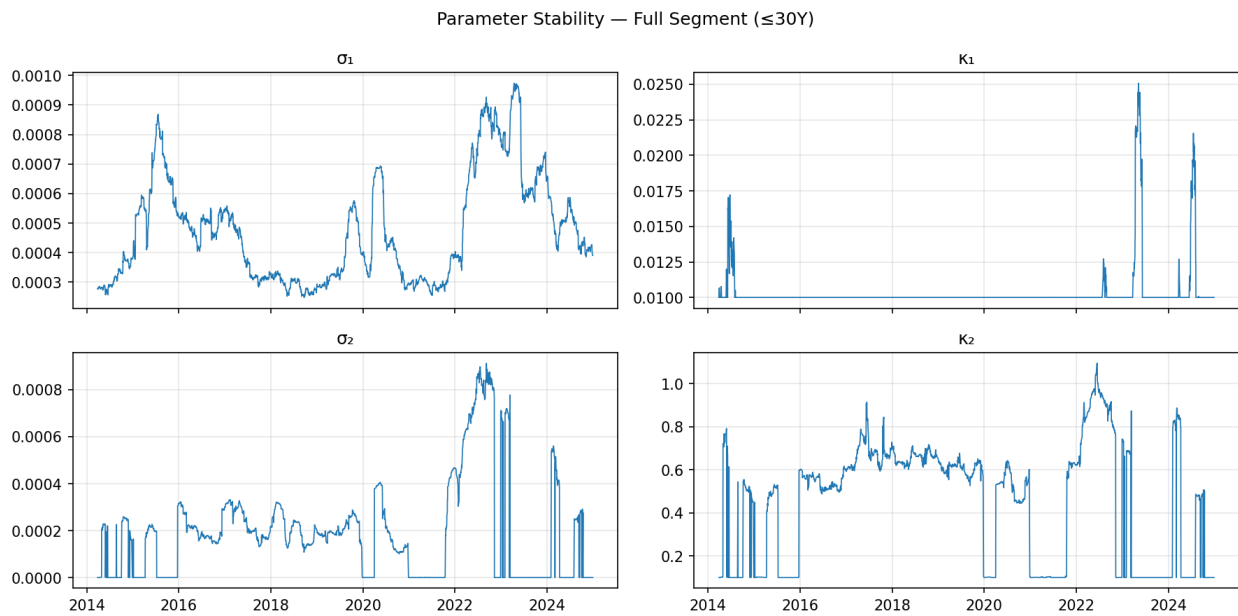


Figure 1: Parameter Stability Across Segments and Dates

6.5 5.5 Three-Factor Comparison

The two-factor rejection could reflect either model inadequacy (a missing factor creates spurious inconsistency) or structural inconsistency (the yield curve segments are genuinely governed by different risk prices). To discriminate, we repeat the test with the three-factor specification and compare the λ gaps.

If the rejection is purely a two-factor artefact, the gaps should vanish uniformly when the curvature factor absorbs the missing cross-sectional variation. If the rejection reflects genuine structural differences, the gaps should persist even with the richer model.

The three-factor results reveal a structural split in the data.

Long-end pairs ($\geq 15Y$). For segment pairs within the long end of the curve — 15Y–20Y, 15Y–25Y,

15Y-30Y, 20Y-25Y, 20Y-30Y, 25Y-30Y — the λ_1 gaps shrink to near zero. The curvature factor absorbs the inconsistency entirely. R^2 improves substantially (from 0.84 to 0.94 for the 30Y segment), confirming that the third factor captures genuine cross-sectional variation. For these pairs, the two-factor rejection reflects model inadequacy masquerading as inconsistency.

Short-vs-long pairs. For pairs involving the 5Y segment against segments of 20Y and above, the λ_1 gaps do not shrink. They *widen*, from approximately -0.09 under the two-factor model to -0.25 to -0.41 under the three-factor model. The curvature factor is calibrated with fundamentally different characteristics across these segments: the 5Y segment produces $\kappa_3 \approx 0.47$ (curvature peaking at $\tau \approx 2$ years), while segments of 15Y and above produce $\kappa_3 \approx 1.1$ – 1.6 (peaking at $\tau \approx 0.6$ – 0.9 years). The third factor is doing different jobs in different segments, and the λ_1 gap absorbs the resulting misidentification.

Intermediate pairs. Pairs involving the 10Y segment against long-end segments show slight widening of gaps but less dramatic than the 5Y pairs, consistent with a gradual transition between the two regimes.

Table: Lambda Gaps Across Models (2-factor vs 3-factor specifications)

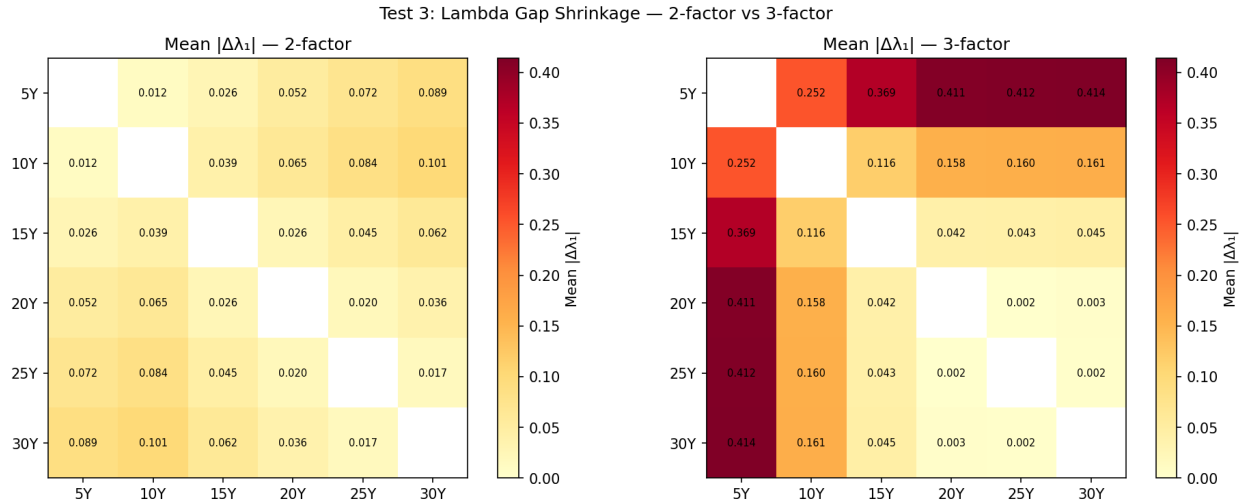


Figure 2: Three-Factor Model: Market Price of Risk Gap Shrinkage by Segment Pair

6.6 5.6 Interpretation

The empirical results demonstrate three layers of discriminating power in the forward-measure consistency test.

Layer 1: The test rejects. Under the two-factor Gaussian HJM, 13 of 15 segment pairs show statistically significant differences in the market price of risk, robust to parameter identification concerns. The rejection is not a calibration artefact — standard goodness-of-fit metrics (R^2) do not flag the inconsistency. This confirms the central methodological claim of the paper: the bottom-up consistency test is non-trivial and has discriminating power that conventional diagnostics lack.

Layer 2: The test distinguishes sources of inconsistency. The three-factor comparison separates two phenomena within the same dataset. The long-end inconsistency ($\geq 15Y$ pairs) vanishes when a curvature factor is added, identifying it as model inadequacy — a missing factor that the two-factor specification cannot accommodate. The short-vs-long inconsistency persists and amplifies, identifying it as structural — the 5Y and 30Y segments of the yield curve are governed by genuinely different risk-pricing dynamics.

Layer 3: The structural inconsistency has economic content. The persistent gap between short and long segments aligns with the preferred-habitat and market-segmentation literature (Modigliani and Sutch 1966, Vayanos and Vila 2021). Short-end bond pricing is dominated by monetary policy expectations, which determine the level and near-term trajectory of short rates. Long-end pricing is dominated by term premia, supply-demand imbalances, and institutional demand from pension funds and insurers. A single global market price of risk cannot reconcile these regimes because they are not generated by the same pricing mechanism.

This observation has implications for the theoretical question posed in Section 3. From the top-down perspective, the existence of a single Q from which all forward measures $\{Q^T\}$ are derived is guaranteed by construction. From the bottom-up perspective, independently calibrated forward measures need not be consistent — and our evidence suggests they are not, once the maturity spectrum is wide enough to span distinct economic regimes. The forward-measure consistency condition is thus empirically testable and economically informative, rather than a mathematical tautology.

Caveat. The empirical test is a joint test of model adequacy and market consistency: rejection means at least one fails. We cannot logically separate them from a single test. The three-factor comparison partially disentangles the two sources, and the economic alignment of the structural gap with known market segmentation provides independent corroboration. Further evidence from alternative model classes (stochastic volatility, quadratic Gaussian), cross-currency replication, and regime-conditional analysis would strengthen the case. We interpret our results as strong evidence suggestive of structural inconsistency, not as proof.

Even the model-inadequacy finding (Layer 2) has practical value. The consistency test detects the need for a curvature factor *before* the two-factor model's R^2 deteriorates to levels that would trigger concern in standard model validation. For practitioners who calibrate term-structure models to subsets of the curve — a common practice in insurance ALM and risk management — this provides a diagnostic that complements conventional goodness-of-fit metrics.

7 6. Mathematical Open Problems

For researchers wishing to pursue the rigorous programme:

1. **Prove or disprove Conjecture 3.2.1** in the Brownian case with deterministic volatility.
2. **Construct an explicit counterexample** where (BF) holds but no consistent Q exists. The diagonal singularity (Section 3.3) identifies the mechanism; rough forward rate models are the

natural territory, though direct evidence for forward rate roughness in bond markets remains to be established.

3. **Characterize Bond-Family NA vs. Herdegen's NAⁿⁱ** in continuous-time bond markets.
4. **Extend to multi-curve:** Bond-Family conditions for each tenor, cross-curve consistency.
5. **Investigate the diagonal singularity** for forward rates with fractional regularity $H < 1/2$ (Filipović 2001, 2009).

Key tools: projective limits of probability measures (Balbás et al. 2002), canonical decomposition of special semimartingales (Jacod-Shiryaev II.2.29), Filipović's SPDE framework for HJM.

8 7. Conclusion

We asked Samuelson's question for bond markets: do observables — tradable bond prices — force the existence of the theoretical construct — the risk-neutral measure Q and the bank account B ? In discrete time, yes: consistency is automatic, and the bank account emerges as a self-financing rolling strategy. In continuous time, the question is open and the answer may be no — the diagonal singularity of the forward rate surface identifies a theoretically grounded mechanism for failure, though an explicit counterexample in a financially relevant setting remains to be constructed.

We tested the question empirically by calibrating local pricing measures independently for different maturity segments of the EUR government bond market. The results are sharp. A two-factor Gaussian HJM model produces statistically significant differences in the market price of risk across 13 of 15 segment pairs — differences that standard goodness-of-fit metrics do not detect. The three-factor comparison separates two distinct phenomena. For long-end pairs, the inconsistency vanishes when a curvature factor is added: model inadequacy, not structural failure. For short-vs-long pairs, the inconsistency persists and amplifies: the 5Y and 30Y segments of the yield curve are governed by genuinely different risk-pricing dynamics, consistent with market segmentation between monetary-policy-driven and term-premium-driven regimes.

The consistency test thus serves a dual role. As a theoretical diagnostic, it gives empirical content to a condition that the standard top-down approach renders tautological. As a practical tool, it detects both model inadequacy and structural inconsistency — and distinguishes between them — using only observable bond prices and standard calibration machinery.

The multi-curve phenomenon of post-2007 is, in our reading, a structural instance of forward measure inconsistency. The bank account emerges not as a starting point but as the output of consistency: it exists if and only if the local pricing measures cohere. Our evidence suggests that within the EUR government bond market, they do not — at least not across the full maturity spectrum.

9 Acknowledgments

The empirical test design, calibration strategy, and three-layer interpretation were developed in extended dialogue with Claude (Anthropic). Claude served as a research collaborator throughout the project — contributing to the formulation of the consistency test, the robustness analysis, the economic interpretation connecting to the preferred-habitat literature, and the structuring of the paper. Data, code, and reproducible results are available at <https://github.com/shaohui1977/bond-family-na>. All errors remain the author’s responsibility.

10 Appendix A: Full Proofs for Section 2

[Complete proofs of Proposition 2.3.1: rolling strategy construction with predictability verification, numeraire invariance lemma, all four equivalences $(NA) \Leftrightarrow (EMM) \Leftrightarrow (Wang) \Leftrightarrow (BF)$, and the consistency derivation. See companion document *section2_final.md*.]

Remark. The change-of-numeraire technique is the standard framework of Geman, El Karoui, and Rochet (1995). See also Musiela and Rutkowski (2005, Chapter 14).

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